

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
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NAME: Randall J. True

eRA COMMONS USER NAME (credential, e.g., agency login):

POSITION TITLE: Founder and CEO, FloodLAMP Biotechnologies, Public Benefit Corporation

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Stanford University, Palo Alto CA	BS w Honors	06/1998	Physics
Stanford University, Palo Alto CA	MS	06/1999	Electrical Engineering

A. Personal Statement

I have the motivation, leadership, knowledge, and connections necessary to solve complex problems and build successful, impactful organizations. I have a broad background in science, engineering, and management, with a focus on the rapid execution of multidisciplinary projects. I have specific expertise and experience in molecular detection technology, including DNA and RNA bioassay development. In addition, I have a wide range of engineering skills spanning hardware, software, and materials. My work has frequently involved design for scale, and I have successfully transferred multiple processes to high volume manufacturing facilities. As the founder of both a biotech startup and an education nonprofit, I'm very familiar with the difficult early stages of projects and the importance of adapting as the organization quickly grows. I encourage open, clear communication and the delegation of high level deliverables in order to efficiently execute a shared plan. After acquisition of my startup by a large public company, I managed a division of 20 FTE's as a VP, interacting with other technical groups, as well as legal, financial, and marketing departments.

Shortly before having our first child in 2018, I followed a lifelong passion for science education and founded a non-profit named Focus on Foundations. The mission of the non-profit is to bring innovative, impactful STEM curricula to a broad range of students by connecting hands-on learning and practical skills with deep theoretical knowledge.

In early March of 2020, I was midway through teaching a 10-week pilot of the Focus on Foundations curriculum when I became aware of the dangers of COVID-19. After initially working on COVID response and hospital modeling with public health professors, I turned my attention to addressing the needs in COVID testing. I founded a Public Benefit Corporation, FloodLAMP Biotechnologies with the mission to improve global health through universal access to molecular testing to find and stop human disease transmission. FloodLAMP is now commercializing a best-in-class rapid molecular mass screening program to address the current pandemic and be better prepared for future ones.

My deepest beliefs are optimistic - that the growth of knowledge is unlimited, along with our capacity to solve problems. The growth of knowledge is also inherently unpredictable and new problems are inevitable. We face many challenging problems in the U.S. and globally, and I personally find meaning in working on solving big problems in education and human health.

B. Positions and Honors

Positions and Employment

2020-2021	Founder and CEO, FloodLAMP Biotechnologies, PBC, San Carlos, CA
2018-2020	Executive Director, Focus on Foundations 501(c) Nonprofit, Portola Valley, CA
2016-2018	Consultant and Advisor to the CEO, Shaper Tools Inc., San Francisco, CA
2008-2010	Vice President of Liquid Arrays, Affymetrix Inc., Santa Clara, CA
2005-2008	Founder & CEO, True Materials, Inc., San Francisco, CA
2001-2004	Senior Engineer, Nanoplex Technologies Inc., Menlo Park, CA
2000-2000	Semiconductor Fabrication Consultant, Intel Corp., Stanford, CA
1998-2000	Process Integration Engineer, Reflectivity Inc., Santa Clara, CA

Other Experience and Professional Memberships

2010-	Entrepreneur, StartUp Advisor/Consultant, Angel Investor, San Francisco, CA
2018-	Career Technical Education Teaching Credential, Santa Clara, CA

Honors

1998	David S. Levine Award for Outstanding Achievement in Physics, Stanford, CA
1998 CA	Rebecca Carrington Prize for scholarship, teaching, & departmental service, Stanford, CA

C. Contributions to Science

Patents

US 8,697,334	Miniaturized microparticles	4-15-2014
US 8,592,136	Methods for producing codes for microparticles	11-26-2013
US 8,460,854	Multiple step printing methods for microbarcodes	6-11-2013
US 8,372,576	Miniaturized microparticles	2-12-2013
US 8,178,278	Miniaturized microparticles	5-15-2012
US 8,168,368	Miniaturized microparticles	5-1-2012
US 8,088,555	Multiple step printing methods for microbarcodes	1-3-2012
US 7,745,092	Multiple step printing methods for microbarcodes	6-29-2010
US 7,745,091	Miniaturized microparticles	6-29-2010
EP 1933817	Encoded Microparticles	3-12-2014

Publications

Microfabricated Templates for the Electrodeposition of Metallic Barcodes for Use in Multiplexed Bioassays, True R.J., Et al, Proceedings of the 26th Annual International Conference on IEEE-EMB (Engineering in Medicine and Biology), Sept 1, 2004.

D. Additional Information: Research Support and/or Scholastic Performance

Resume Attached

Randall J. True

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EXPERIENCE

August 2020 – Present: Founder and CEO, FloodLAMP Biotechnologies PBC San Carlos, CA

- Building up a biotech startup with the mission to improve global health and resiliency through universal access to rapid molecular testing.
- Delivering an integrated RT-LAMP COVID testing program combining distributed sample collection with sample pooling to enable frequent, inexpensive screening of schools and other interacting populations.
- Focusing on public good mission through shared protocols, workflows, operational procedures, and materials in an effort to accelerate this important new disease control modality.

September 2018 – July 2020: STEM Education Development Portola Valley, CA

- *Focus on Foundations* – 501(c)3 nonprofit for public charity, founded in Dec 2018 to develop next-gen STEM education curricula. First major project is a hands-on course based on building an 8-bit CPU from scratch, learning the fundamentals of electronics and computation, as well as intersections with math and physics.
- California CTE Teaching Credential - Volunteer High School CTE Teacher at SVCTE in San Jose, was midway through 10 week pilot of 8-bit computer curriculum when the school closed for COVID.
- Education Related Volunteering
 - VEX Robotics Club mentor at *Corte Madera Middle School* in Portola Valley, CA
 - Panelist/judge for the FlexFactor Education Program at *NextFlex* (Advanced Manufacturing Institute in San Jose CA)
- Attended EdSurge Fusion Conference in Oct 2018

July 2010 – Present: Entrepreneur, StartUp Advisor/Consultant, Angel Investor San Francisco, CA

Invested and Advised/Consulted in projects in a variety of fields:

- Software Service – *NAT AI* (Outdoor Experience planning and automation, SF CA)
- Hardware Devices - *Flow Labs Inc.* (GPS tracking device for surfing, SF CA)
- Biotech - *Purigen Inc.* (DNA separation technology, Palo Alto CA)
- Restaurant/Food - *Spike Kit* (quick serve Asian cuisine, CA),
- Independent Film – Executive Producer of *About Cherry* (selected for Berlin Intl Film Festival,) and *American Milkshake* (selected for Sundance Film Festival)
- Tourism and Cycling - *Streets of San Francisco* (bike tour and rental company)
- Real Estate – *Cigar Factory Lofts* (7 unit condo development in Oakland, CA)

Cooking: Stage at *Sous Beurre Kitchen* (Mar 2012)

Travel: Southern Patagonia, Argentina, Buenos Aires, Medellin, Colombia, Uruguay, Brazil

India, Taiwan, Thailand, Germany, Italy, France, Spain, Morocco

California, MidWest US, Southwest US, Maui and Hawaii, Canada, New Zealand, Australia

Recent Athletics: USCF Category 2 Cycling Road Racing License (2015)

2015 NorCal State Cycling Time Trial, 2nd place in Master's Category, age 35-40

Jan 2016 – Jan 2018: Consultant and Advisor to the CEO, Shaper Tools Inc San Francisco, CA

- Led strategy for Series A investment by Festool, who ended acquiring Shaper Tools
- Developed IP strategy
- Beta tested early prototypes or Origin computer vision CNC router

June 2008 – July 2010: Vice President of Liquid Arrays, Affymetrix Inc.

Santa Clara, CA

- Lead the integration and development of the Multiplexed Microbarcode Assay Platform acquired from startup True Materials Inc.
- Grew the interdisciplinary Liquid Array team to 20+
- Developed technology through pilot manufacturing stage

Feb 2005 – June 2008: Founder & CEO, True Materials, Inc.

San Francisco, CA

- Founded biotech startup based on a microbarcode platform for multiplexed bioassays.
- Developed applications in genetic analysis and molecular diagnostics.
- Incubated in first cohort of companies at UCSF/QB3 Garage at Mission Bay.
- Acquired in 2008 by Affymetrix, then a public microarray company, now a part of Thermo Fisher

Jan 2000 – Oct 2005: Microfabrication Consultant

Stanford, CA

- Microfabrication related services, specialization in Micro Electro Mechanical Systems (MEMS).
- Clients include startups, large corporations, and academic research groups.
- Services based at the Stanford Nanofabrication Facility.
- Fabrication expertise in photolithography (stepper and contact), plasma etching, chemical vapor deposition, evaporation, sputtering, and wet processing.

Sept 2001 - Nov 2004: Senior Engineer, Nanoplex Technologies, Inc.

Menlo Park, CA

- Managed the development of a new class of metallic nanoparticles – lithographic Nanobarcodes.
- Designed and fabricated templates using semiconductor processes including polymer deposition, i-line photolithography and deep reactive ion etching.
- Integrated all steps of the project: template fabrication at the Stanford Nanofabrication Facility, electrodeposition, characterization using SEM and optical microscopy, development of software algorithms for Nanobarcode identification, and optimization of the bioassay protocol.
- Coordinated external fabrication services.
- Initiated and managed international x-ray lithography collaboration.

Mar 2000 - Dec 2000: Semiconductor Fabrication Consultant, Intel Corp.

Stanford, CA

- Designed and implemented semiconductor processes for a novel optoelectronic device.
- Fabricated passive waveguides and active optical structures on SOI substrates.
- Interacted with Intel device design and testing engineers.
- Trained other Intel employees in cleanroom and processing procedures.

Jun 1998 - Jan 2000: Process Integration Engineer, Reflectivity, Inc.

Santa Clara, CA

- Contributed to development of Reflectivity's semiconductor-based/MEMS microdisplay technology.
- Was the first employee of Reflectivity, mentored by founders from pre-seed thru 2nd round funding.
- Fabricated first and second generations of working prototypes including color XGA unit.
- Tasks: device design, mask layout, process development (deposition, lithography, and etching), testing.
- Transferred process to manufacturing partner in Taiwan, worked on-site in foundry for 6 weeks.
- Developed CCD-based optical characterization system for completed microdisplays.
- Assisted in the development of a XeF₂ gas-phase etching system.
- Researched patents of competing technologies and wrote patent disclosures.
- Attended industry conferences.
- Trained process engineers.

PATENTS

US 8,697,334 Miniaturized microparticles 4-15-2014

US 8,592,136 Methods for producing codes for microparticles 11-26-2013

US 8,460,854 Multiple step printing methods for microbarcodes 6-11-2013

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EP 1933817 Encoded Microparticles 3-12-2014

PUBLICATIONS

[*Loop-Mediated Isothermal Amplification \(LAMP\) Detection of SARS-CoV-2 and Myriad Other Applications*](#), K. Moore, R. True, A. Wyllie, C. Mason, et al., Journal of Biomolecular Techniques. Volume 32 (3); 2021 September, pg. 89.

Microfabricated Templates for the Electrodeposition of Metallic Barcodes for Use in Multiplexed Bioassays, True R.J., et al., Proceedings of the 26th Annual International Conference on IEEE-EMB (Engineering in Medicine and Biology), Sept 1, 2004.

EDUCATION

Stanford University	M.S. in Electrical Engineering	June 1999
Stanford University	B.S. in Physics with Honors	June 1998

Mar 1996 - Jun 1998: Undergraduate Researcher, Stanford Physics Department Stanford, CA
Lead project to develop a superconducting transistor in low-dimensional GaAs/AlGaAs heterostructures. Fabricated devices in Stanford's National Nanofabrication Facility using electron beam lithography and ECR plasma etching, measured transport through superconductor-coupled quantum dots, Supervised and trained other undergraduate researchers.

Research Grants Awarded:

Bing Fellowship - \$3,000 (1996), Center for Materials Research Summer Fellowship - \$3,500 (1997), Undergraduate Research Opportunities Major Grant - \$2,500 (1997)
NSF User Grants through the Stanford Nanofabrication Facility - \$14,000 (5/1996-6/1998)

EE Coursework: Computer Organization and Design, Programming Methodology, Applied Superconductivity, Physics for Solid State Electronics, Semiconductor Optoelectronic Devices, Solid-State Sensors and Actuators, Semiconductor Materials.

Physics Coursework: Classical Mechanics, Analog Electronics, Electricity and Magnetism, Quantum Mechanics, Statistical Mechanics, Solid State Physics.

Awards: David S. Levine Award for Outstanding Achievement in Physics.

Rebecca Carrington Prize for scholarship, teaching and departmental service

ATHLETICS

Cycling

United States Cycling Federation: Category 2 on Road (2004, 2015), Category 2 on Track (2003-2004)

2015 NorCal State Time Trial, 2nd Place Masters 35-39 age group

2004 Northern California Track District Championships: 4km Pursuit –2nd place

Major Rides include Haleakala 12 times (10K ft ascent), Death Ride, San Juan Huts Mountain Tour

Mountaineering/Climbing

Traverse of Southern Patagonia Ice Field – Feb 2013

Ascent Mt Shasta, CA (14,180ft) - 1997

Red Rocks in Nevada CA, Lovers Leap

Technical Canyoneering: Zion NP and Lake Powell AZ